

AVNEET AURORA

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EDUCATION

University of Delaware, Honors College — Newark, DE

Expected May 2029

B.S. Computer Engineering — **Junior by Credit** | GPA: 3.77 | Dean's List

Coursework: Data Structures & Algorithms, Machine Learning, Software Engineering, Digital Logic Design, Linear Algebra II, Calc III

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript/TypeScript, SQL, VHDL

ML & Data: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib

Backend & Cloud: FastAPI, Node.js, REST API design, PostgreSQL, AWS (EC2, RDS), CI/CD, Linux/Bash, Git

Frontend & Mobile: React, Next.js, React Native (Expo), HTML/CSS

EXPERIENCE

Undergraduate Research Assistant — Vertically Integrated Projects (Scientific Computing) **Feb 2026 – Present**

- Extended peer-reviewed 2D Lattice Boltzmann fluid-simulation research to 3D (D3Q27 lattice) by replacing the analytical BGK collision operator with a learned neural network, enabling data-driven simulation of turbulent flows
- Improved model generalization and shrank the parameter search space by engineering TensorFlow/NumPy training pipelines that enforce physical lattice-symmetry constraints on the learned operator
- Achieved mass-conservation error at machine precision ($\sim 1e-15$) by validating the 3D model against analytical solutions, confirming the operator preserves the underlying physics; presented results to the research team

Undergraduate Research Assistant — Winter Fellows, Electrical & Computer Engineering **Jul 2025 – Feb 2026**

- Raised wildfire-ignition validation accuracy by 10–20% over a baseline stuck at $\sim 50\%$ by re-architecting a PyTorch CNN (ReLU, batch normalization, regression head), correcting the data pipeline, and expanding the training set
- Selected for the merit-based Winter Fellows award to extend 3D ignition-estimation research from airborne to satellite LiDAR point clouds; presented findings at the Winter Undergraduate Research Showcase
- Reduced per-plot preprocessing from a manual multi-step workflow to a single scripted run by automating Python pipelines for point-cloud normalization, forest-metric computation, and heatmap visualization

Technology Intern — Code Differently × Buccini Pollin Group, Wilmington, DE

Jun 2024 – Aug 2024

- Analyzed real-estate project financials and presented a data-driven amenity investment proposal directly to company founders; awarded a \$1,000 Metropolitan Urban League scholarship

PROJECTS

Decora — Full-Stack AI Interior Design Platform (decora.club) | Next.js, TypeScript, Supabase, Stripe **Aug – Nov 2025**

- Shipped a production AI room-redesign platform end to end: serverless REST API on Vercel, PostgreSQL backend, image object storage, and Git-based CI/CD, delivered through an agile, iterative release workflow
- Integrated live Stripe payment processing with user authentication and a credit-billing transaction ledger schema

SquatBuddy — Real-Time AI Form Coach (HenHacks 2026) | PyTorch, OpenCV, LSTM **Mar 2026**

- Engineered a real-time video pipeline that extracts joint keypoints with OpenCV and detects squat-form faults frame by frame using an LSTM autoencoder, delivering joint-specific biomechanical feedback across the full kinetic chain

TaskAI — AI Study Scheduler | Winner, Best Technical — UD CIS Honors Hackathon | Python, Gemini API **Dec 2025**

- Won Best Technical for a web app integrating Google Gemini to generate constraint-aware study schedules; designed the UI and scheduling logic in a 24-hour build

LEADERSHIP & AWARDS

President, DevLab Club — University of Delaware

Feb 2026 – Present

- Grew weekly attendance 4x (from 3–5 to 20+ members) by designing and teaching a full-stack curriculum (Git workflows, REST API design, OOP, LLM-assisted development); mentor members through architecture, debugging, and code review

Awards: Winter Fellows Research Award (2025) • Best Technical, UD CIS Honors Hackathon (2025) • Metropolitan Urban League Scholarship (2024)